Marlenne Esparza

maesparza26@gmail.com • 682-402-4149
https://maesparza26.github.io/ • Fort Worth, TX

EDUCATION

Texas A&M University – College Station, TX *May 2026*
Bachelor of Science in Multidisciplinary Engineering Technology – Mechatronics ***GPA***: 3.20/4.00
Minors: Embedded Systems Integration, Mathematics

SKILLS

Robotics & Automation: ROS (RViz, MoveIt 2), UR Cobot Programming (Polyscope)
Programming: Python, C/C++, Assembly, Kotlin, React/Next.js, Linux
Instrumentation & Testing: Oscilloscope, Multimeter, Signal Generator, AC/DC Power Supply
Hardware & Electronics: PCB Design, KiCad, Multisim, Quartus, LiDAR, Node-RED, SPI, I2C, UART
Modeling & Simulation: SolidWorks, Autodesk Fusion 360, Onshape, MATLAB, Simulink
Manufacturing & Machining: Mill, Lathe, 3D Printing

PROJECTS

Bag Valve Mask Monitoring System *August 2025 – May 2026*

- Designed and fabricated a sensor-integrated attachment for a bag valve mask to monitor patient respiratory metrics.
- Integrated an embedded system to acquire and process real-time sensor data, interfacing pressure and temperature sensors with Bluetooth communication modules.
- Built a web-based dashboard to visualize real-time data with dynamic graphs, alerts, and system status indicators for clinical monitoring.

PID-Controlled Ball Balancing Platform *November 2025*

- Designed a closed-loop PID control system to stabilize a steel ball to center of a two-axis tilting platform.
- Modeled system dynamics and tuned PID gains to minimize overshoot, settling time and steady-state error.
- Validated controller performance with MATLAB Simulink and physical testing.

Stack & Sort Robot *April 2025*

- Prototyped and assembled an autonomous robot capable of detecting, grasping, and stacking colored blocks.
- Implemented motor control using PWM signals to drive stepper and servo motors via Raspberry Pi GPIO.
- Integrated computer vision using HSV color space detection for real-time object recognition and sorting.

Hanging Crane with 2-Stage Pendulum *November 2024*

- Constructed a hanging crane prototype incorporating a two-stage pendulum to analyze system dynamics and control behavior.
- Developed a simulation model in MATLAB/Simulink and validated system performance by comparing simulated results with experimental prototype data.

EXPERIENCE

Texas A&M University – College Station, TX *June 2025 - July 2025*
Undergraduate Research, Department of Engineering Technology and Industrial Distribution

- Redesigned and fabricated an open-source robotic hand, improving mechanical functionality and modularity.
- Reverse-engineered existing CAD models to develop modifiable design files in SolidWorks, enabling design customization and iterative prototyping.

Sophomores Leading on Promoting Equality – College Station, TX
Service and Outreach Director *March 2025 – May 2026*

- Collaborated with the Fundraising Committee to support philanthropy efforts benefiting *Health For All*, including organizing and managing an annual fundraising event.
- Partnered with local organizations and charities to expand outreach and increase community engagement.

Robomaster's Engineer Team – College Station, TX
Team Member *January 2023 - January 2024*

- Designed and integrated mechanical subsystems for competition robots, including gripper and conveyor mechanisms for object manipulation and transport.